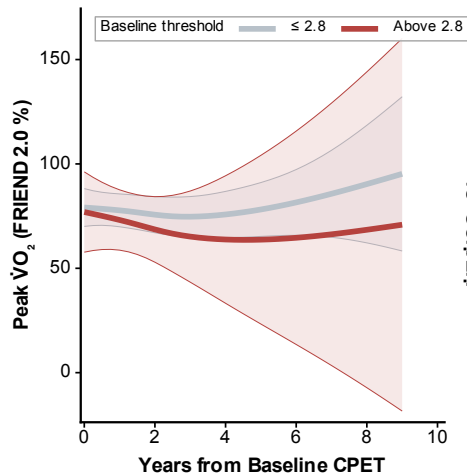


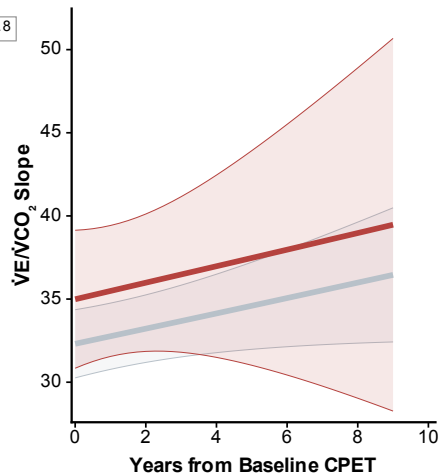
(A) Peak $\dot{V}O_2$

Baseline TRVmax ≤ 2.8 m/s versus above 2.8 m/s



(B) $\dot{V}E/\dot{V}CO_2$ slope

Baseline TRVmax ≤ 2.8 m/s versus above 2.8 m/s



(C) Longitudinal Model Summary

Model component	edf	F statistic	P value
Peak $\dot{V}O_2$			
Time smooth	1.84	0.63	0.532
Threshold group	—	0.71	0.401
Time x threshold group	1.00	0.21	0.652
$\dot{V}E/\dot{V}CO_2$ slope			
Time smooth	1.02	4.10	0.043
Threshold group	—	1.61	0.206
Time x threshold group	1.01	0.00	0.977

Supplemental Figure. Parameter-specific longitudinal cardiopulmonary trajectories stratified by binary baseline TRVmax (m/s) status. **(A)** GAMM-derived Peak $\dot{V}O_2$ trajectories. **(B)** GAMM-derived $\dot{V}E/\dot{V}CO_2$ slope trajectories. **(C)** Summary of the key status and interaction terms for both models. Baseline threshold groups were defined as TRVmax ≤ 2.8 m/s versus above 2.8 m/s. Longitudinal models were adjusted for age, sex, and BMI.